

Do Modern Deep Learning Architectures Disproportionately Benefit BCI-Inefficient Users? An Empirical Analysis on Motor Imagery EEG Classification

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Abstract

Brain-computer interfaces (BCIs) based on motor imagery electroencephalography (EEG) offer transformative potential for assistive technologies, yet an estimated 15–30% of users cannot effectively operate them — a phenomenon termed BCI inefficiency. Prior work has shown that convolutional neural networks improve classification for inefficient users relative to traditional signal processing in binary tasks [15], but this has not been extended to multi-class settings or newer architectures. We extend this investigation to 4-class motor imagery using BCI Competition IV Dataset 2a [16], comparing five approaches: Filter Bank Common Spatial Pattern (FBCSP) [3], EEGNet [11], ATCNet [2], a Mamba-based selective state space model [7, 8], and a Bootstrap Your Own Latent (BYOL) [6] self-supervised approach with leave-one-subject-out cross-subject pretraining. High and low performer groups are defined via median split on FBCSP kappa, and we analyze whether gains over the baseline are disproportionately larger for low performers, with and without Segmentation and Recombination (S&R) augmentation [14]. Spectral analysis confirms that low performers exhibit absent mu/beta event-related desynchronization over motor cortex, providing neurophysiological grounding for their classification difficulty. EEGNet and ATCNet consistently trend toward favoring low performers, while Mamba underperforms FBCSP for all subjects. SSL-BYOL achieves the strongest overall performance with the largest individual gains for subjects with most absent ERD, and an ablation study reveals heterogeneity in pretraining contributions across low-performing subjects. These results constitute the first systematic analysis of BCI inefficiency across this model hierarchy in 4-class classification and the first investigation of whether cross-subject SSL pretraining disproportionately benefits BCI-inefficient users.

1. Problem Description and Motivation

Brain-computer interfaces based on motor imagery EEG have emerged as a promising technology for individuals with severe motor disabilities, enabling direct communication between the brain and external devices such as prosthetic limbs, wheelchairs, and augmentative communication systems without requiring any physical movement. The core signal underlying most motor imagery BCIs is event-related desynchronization (ERD) — a reduction in mu (8–12 Hz) and beta (13–30 Hz) oscillatory power over the sensorimotor cortex that occurs when a user imagines performing a movement. Classifiers trained to detect these patterns can decode motor intent with sufficient accuracy for practical control in favorable conditions.

A persistent and clinically significant obstacle is BCI inefficiency: an estimated 15–30% of users are unable to produce ERD patterns robustly enough for a classifier to exceed 70% accuracy, even after extensive training [4, 1]. This is not merely an accuracy problem — it is an equity problem. The patients who most urgently need BCI technology, including individuals with ALS, stroke survivors, and upper-limb amputees, are often precisely those who are harder to classify, whether due to neurological disruption, limited motor cortex plasticity, or natural inter-individual variability in sensorimotor rhythm generation. Critically, Ahn and Jun [1] demonstrated that high resting-state theta and low alpha power may predict poor BCI performance, suggesting that BCI inefficiency has identifiable neurophysiological correlates rather than being purely random variability.

Recent deep learning models have dramatically improved mean accuracy on benchmark datasets such as BCI Competition IV Dataset 2a [16], but nearly all published work reports average accuracy across subjects without examining whether these gains are distributed equitably across the performance spectrum. Nakanishi et al. [15] provided the first direct evidence that CNNs disproportionately benefit inefficient users in a 2-class online BCI paradigm, with improvements of up to 28% for the lowest performers relative to a CSP+LDA baseline. However, that work was limited to binary classification with online accuracy scores, leaving open whether the finding extends to the more clinically relevant 4-class setting, to offline evaluation protocols, and

to more recent architectures including temporal CNNs, state space models, and self-supervised pretraining. Answering these questions has direct implications for model selection in clinical deployment, where the goal is not maximizing mean accuracy across a population but ensuring equitable performance for the full range of potential users.

2. Research Questions

This work addresses four research questions spanning both the architectural and data dimensions of BCI inefficiency.

RQ1: Does EEGNet improve classification accuracy for BCI-inefficient subjects more than BCI-efficient subjects, relative to the FBCSP baseline? This extends the binary-task finding of Nakanishi et al. [15] to 4-class motor imagery on BCI IV-2a.

RQ2: Does ATCNet further amplify this effect relative to EEGNet, and does a Mamba-based state space model [7] provide additional benefit or exacerbate the gap? ATCNet combines convolutional spatial filtering with multi-head self-attention and a temporal convolutional network, while Mamba’s selective state space mechanism adaptively retains or discards temporal information — a property that could in principle capture weak or irregular ERD patterns, but which comes with a higher parameter count relative to training data.

RQ3: Does Segmentation and Recombination (S&R) data augmentation [14] differentially benefit low-performing subjects, or does it shift accuracy uniformly across both groups?

RQ4: Does cross-subject BYOL self-supervised pretraining [6] disproportionately benefit BCI-inefficient subjects? We implement a leave-one-subject-out (LOSO) pretraining design in which the BYOL encoder is trained on unlabeled EEG from the other eight subjects before fine-tuning on the target subject’s labeled data. This design is motivated directly by the spectral finding that low performers exhibit absent mu/beta ERD — if canonical ERD is not present in a subject’s training data, a model pretrained on diverse EEG representations from other subjects may learn to detect alternative neural patterns that supervised training alone cannot discover.

3. Proposed Solution

3.1. Dataset and Preprocessing

All experiments use BCI Competition IV Dataset 2a [16, 5], which contains EEG recordings from 9 subjects performing 4 motor imagery tasks: left hand (event 769), right hand (770), feet (771), and tongue (772). Each subject has two sessions: session 1 (A0#T.npz, 288 training trials, 72 per class) and session 2 (A0#E.npz + A0#E.mat, 288 evaluation trials with true labels). All 22 EEG channels are used; 3 EOG channels are discarded. The session 1 / session 2 split is used throughout, consistent with the original competition protocol and the published literature.

FBCSP operates at the original 250 Hz sampling rate with 1000-sample (4-second) epochs, consistent with Ang et al. [3]. All neural network models apply 5th-order Butterworth bandpass filtering (4–40 Hz), downsample signals to 128 Hz via polyphase resampling (`resample_poly`, `up=64`, `down=125`), and extract 512-sample (4-second) epochs. Per-channel StandardScaler normalization is applied (fit on training session, applied to evaluation session), following the preprocessing fixes in the `amrzhhd/EEGNet` implementation.

3.2. Models

FBCSP (traditional baseline): Filter Bank Common Spatial Pattern with LDA, one-vs-rest multiclass extension [3], implemented using the `jesus-333/FBCSP-Python` codebase [18]. Filter bank: 9 Butterworth bands from 4–40 Hz in 4 Hz steps, order 3. Features selected by MIBIF (top 4 per band). Validated against published kappa scores in Table 2 of Ang et al. [3].

EEGNet (CNN baseline): Compact convolutional neural network for EEG [11], configured with temporal kernel length 64 (500 ms at 128 Hz), $F1=16$, $D=2$, $F2=32$, dropout 0.5. Implemented from `amrzhhd/EEGNet` (PyTorch), which includes per-channel normalization not present in the original paper. Trained with Adam ($\text{lr}=0.001$), 300 epochs, batch size 32.

ATCNet (temporal CNN baseline): Attention Temporal Convolutional Network [2], instantiated via `braindecode` with `n_chans=22`, `n_outputs=4`, `n_times=512`, `sfreq=128`. ATCNet combines a convolutional projection block, multi-head self-attention, and a dilated causal TCN. A wrapper layer squeezes the extra channel dimension to match the (batch, 22, 512) input expected by ATCNet. Trained identically to EEGNet.

MI-Mamba (state space model): Hybrid CNN-Mamba architecture adapted from Guo et al. [8]. A single Conv2d(1, 32) layer collapses all 22 electrodes to $d_{\text{model}}=32$ spatial features, followed by a Mamba block ($d_{\text{model}}=32$, $d_{\text{state}}=16$, $d_{\text{conv}}=4$, $\text{expand}=2$) [7] for global temporal processing, dropout 0.5, and a linear classifier. The Mamba block uses approximately $3 \times \text{expand} \times d_{\text{model}}^2$ parameters. Multiple configurations were explored during development: $d_{\text{model}}=64$ led to severe overfitting (near-zero training loss by epoch 100, test accuracy near chance); the final $d_{\text{model}}=32$ with dropout was the most stable. Trained with Adam ($\text{lr}=0.001$), 500 epochs, batch size 32.

SSL-BYOL (self-supervised): Bootstrap Your Own Latent [6] with leave-one-subject-out pretraining. The online network uses an EEGNet backbone with the final classification layer replaced by a projection head (Linear(128→256)–BN–ReLU–Linear(256→256)) and a predictor head (Linear(256→128)–BN–ReLU–Linear(128→256)). The target network is an exponential moving average (momentum=0.996) of the online network. EEG-specific augmentations applied during pretraining: mild Gaussian noise (3% of signal std), random temporal masking (0.2–0.6s window), and amplitude scaling (uniform $\pm 20\%$). Pretraining: 300 epochs, batch size 64, Adam $\text{lr}=0.001$, $\text{weight_decay}=1\text{e-}4$, on 2304 unlabeled trials from 8 subjects. Fine-tuning: online backbone with new Linear(128→4) head, 300 epochs, same optimizer, on target subject’s 288 labeled trials.

3.3. Group Definition and Analysis

Subject performance groups are defined via a median split of per-subject FBCSP accuracy across the 9 subjects: **high performers** (subjects 01, 03, 07, 08, 09) and **low performers** (subjects 02, 04, 05, 06). This split is consistent with published per-subject rankings in the literature [9]. The core analysis computes $\Delta\text{Acc} = \text{Acc}_{\text{model}} - \text{Acc}_{\text{FBCSP}}$ per subject and compares the mean ΔAcc between groups using a Mann-Whitney U test (appropriate for $n=4$ vs $n=5$; minimum achievable $p=0.095$). Results are interpreted as exploratory given the mathematical constraint on statistical power.

3.4. Data Augmentation

Segmentation and Recombination (S&R) [14] is applied to the training session only. Each 4-second trial (512 samples at 128 Hz) is split into two equal 256-sample segments at the 2-second midpoint. Synthetic trials are created by randomly selecting each segment from a different same-class donor trial, expanding from 288 to 864 training trials per subject ($3\times$). Augmentation is applied before tensor conversion; the evaluation session is never augmented.

4. Relation to Prior Work

The most directly related work is Nakanishi et al. [15], which demonstrated that CNNs outperform CSP+LDA for inefficient users in a 2-class online BCI paradigm, with gains of up to 28% for the lowest performers. We extend this to 4-class offline classification and a broader model hierarchy. The FBCSP baseline on BCI IV-2a is established by Ang et al. [3], who reported per-subject kappa values for all 9 subjects; EEGNet has been evaluated on this dataset by Lawhern et al. [11] and subsequent work. ATCNet on BCI IV-2a was introduced by Altaheri et al. [2], reporting 75.96% mean accuracy. Mamba-based models for motor imagery EEG are very recent: MI-Mamba [8] reports 80.59% on BCI IV-2a; STMambaNet [17] reports similar performance. Neither work examines per-subject results through the BCI inefficiency lens.

The BCI illiteracy phenomenon is documented by Blankertz et al. [4], who reframed poor BCI performance as a system failure rather than a user failure. Ahn and Jun [1] identified neurophysiological correlates of BCI inefficiency, showing that high resting theta and low alpha power predict poor performance. Kim et al. [9] directly address BCI illiteracy on BCI IV-2a using a semantic style transfer approach, identifying the same low-performing subjects as our median split; however, they do not perform a systematic comparison across model architectures.

Self-supervised learning for EEG has been explored by several recent works. BENDR [10] established the pretrain-then-finetune paradigm for EEG. Li et al. [12] apply contrastive SSL specifically for cross-subject motor imagery on BCI IV-2a. Grill et al. [6] introduced BYOL as a negative-sample-free alternative to SimCLR, which we adopt because SimCLR’s NT-Xent loss requires large batch sizes for adequate negative pairs — empirically, SimCLR failed to converge below a loss of 3.9 on our 2304-trial pretraining set, while BYOL converged to a cosine similarity of approximately 0.90. No prior work has asked whether SSL pretraining disproportionately benefits BCI-inefficient subjects, which is the novel contribution of RQ4.

Data augmentation for EEG motor imagery has been studied in Zhao et al. [20]. S&R was introduced by Lotte et al. [14] and validated by Roy et al. as outperforming Gaussian noise, sliding window, and window warping for EEG classification. We adopt S&R over GAN-based methods [19, 13] because its deterministic, interpretable mechanism — recombining segments from same-class trials — is more appropriate for studying differential effects across subject groups, and because GAN training on only 72 trials per class risks mode collapse for the weakest subjects.

5. Experimental Results

5.1. Supervised Model Comparison

Table 1 shows per-subject accuracy for all models in original (no augmentation) and S&R augmented conditions. FBCSP achieves a mean accuracy of 63.4%, consistent with published kappa values from Ang et al. [3]. EEGNet (71.4%) and ATCNet (69.1%) both improve substantially over FBCSP. Mamba (51.2%) underperforms FBCSP for 7 of 9 subjects, failing to generalize on 288 training trials. The subject-level ranking is consistent across all models: subjects 03, 08, and 01 are consistently top performers; subjects 02, 05, and 06 are consistently the weakest.

Table 1: Per-subject accuracy for all models. High performers (H): 01, 03, 07, 08, 09. Low performers (L): 02, 04, 05, 06. Best result per subject in **bold**.

Model	S01	S02	S03	S04	S05	S06	S07	S08	S09
Group	H	L	H	L	L	L	H	H	H
FBCSP	0.677	0.542	0.806	0.587	0.507	0.424	0.771	0.719	0.677
EEGNet	0.823	0.583	0.854	0.670	0.632	0.583	0.719	0.819	0.743
ATCNet	0.792	0.545	0.882	0.597	0.677	0.601	0.757	0.813	0.774
Mamba	0.604	0.354	0.625	0.458	0.278	0.299	0.490	0.705	0.597
EEGNet+S&R	0.813	0.608	0.889	0.642	0.712	0.601	0.802	0.865	0.823
ATCNet+S&R	0.802	0.604	0.840	0.580	0.715	0.594	0.760	0.840	0.809
Mamba+S&R	0.667	0.347	0.688	0.326	0.319	0.410	0.559	0.708	0.611
SSL-BYOL	0.778	0.549	0.851	0.569	0.587	0.566	0.795	0.788	0.705
SSL-BYOL+S&R	0.785	0.545	0.865	0.667	0.684	0.576	0.868	0.802	0.819

Table 2 presents the BCI illiteracy group analysis. The gap column (Low–High) is the key quantity: positive values indicate the model disproportionately helped low performers relative to FBCSP. EEGNet and ATCNet both show positive gaps without augmentation (+0.040 and +0.023), consistent with the Nakanishi et al. hypothesis. Mamba shows a negative gap in all conditions, meaning it hurts low performers more than high performers. SSL-BYOL+S&R achieves the largest absolute ΔAcc for both groups while maintaining a near-zero gap (+0.005), indicating equitable benefit across the performance spectrum.

Table 2: BCI illiteracy group analysis. ΔAcc = model accuracy – FBCSP accuracy. Gap = Low–High ΔAcc (positive = model disproportionately helps low performers). None reach $p < 0.05$ due to the mathematical constraint of $n=4$ vs $n=5$ (minimum achievable $p=0.095$).

Model	High ΔAcc	Low ΔAcc	Gap	p
EEGNet (original)	+0.062	+0.102	+0.040	0.556
EEGNet + S&R	+0.108	+0.126	+0.018	0.730
ATCNet (original)	+0.046	+0.069	+0.023	0.905
ATCNet + S&R	+0.044	+0.111	+0.067	0.556
Mamba (original)	−0.102	−0.148	−0.046	0.286
Mamba + S&R	−0.083	−0.164	−0.081	0.286
SSL-BYOL (no aug)	+0.044	+0.048	+0.004	1.000
SSL-BYOL + S&R	+0.098	+0.103	+0.005	0.905

5.2. Architectural Analysis: Why ATCNet Learns and Mamba Fails

The performance gap between ATCNet and Mamba is explained by the mismatch between model capacity and available training data. ATCNet succeeds because its three-component architecture provides strong EEG-specific inductive bias: a convolutional projection block performs spatial filtering analogous to CSP, a multi-head self-attention module captures global temporal context across the 512-sample trial, and a dilated causal TCN models sequential dependencies. All three components are designed for fixed-length sequences with strong spatial structure — precisely the properties of EEG motor imagery.

Mamba’s selective state space mechanism was originally designed for very long sequences (thousands to millions of tokens) in NLP, where its linear-complexity advantage over quadratic attention is meaningful [7]. On a 512-sample EEG trial, this advantage is irrelevant. More critically, the Mamba block uses approximately $3 \times \text{expand} \times d_{\text{model}}^2$ parameters — with $d_{\text{model}}=32$ and $\text{expand}=2$, approximately 6,144 parameters in the Mamba layer alone — representing significant overcapacity relative to 288 training trials. Our debug loss curves confirm this: MI-Mamba reached near-zero training loss by epoch 100 (complete memorization of training data) while achieving only 67% test accuracy on the easiest subject (S03), indicating overfitting rather than generalization. We explored multiple configurations to mitigate this: $d_{\text{model}}=64$ collapsed to chance-level performance; learning rates of $1e-4$ underfit; $\text{weight_decay}=1e-2$ destroyed learning entirely (all subjects near 25% chance); the final configuration ($d_{\text{model}}=32$, $\text{weight_decay}=1e-4$, $\text{dropout}=0.5$) represented the best balance achievable within the small-data constraint. Gradient clipping ($\text{max_norm}=1.0$) produced no meaningful improvement (mean 0.486). The fundamental limitation is architectural: Mamba lacks the spatial inductive bias that makes EEGNet and ATCNet robust to small training sets.

5.3. Confusion Matrix Analysis

Figure 1 shows aggregate confusion matrices across all 9 subjects for each model, without augmentation (top) and with S&R augmentation (bottom).

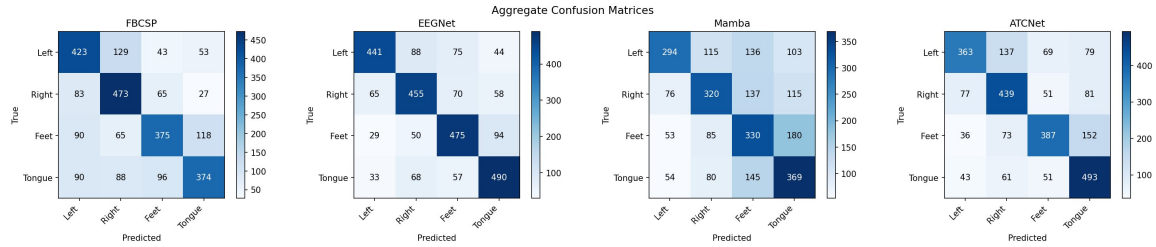


Figure 1: Aggregate confusion matrices across all 9 subjects without augmentation. Each cell shows raw trial counts summed over all subjects.

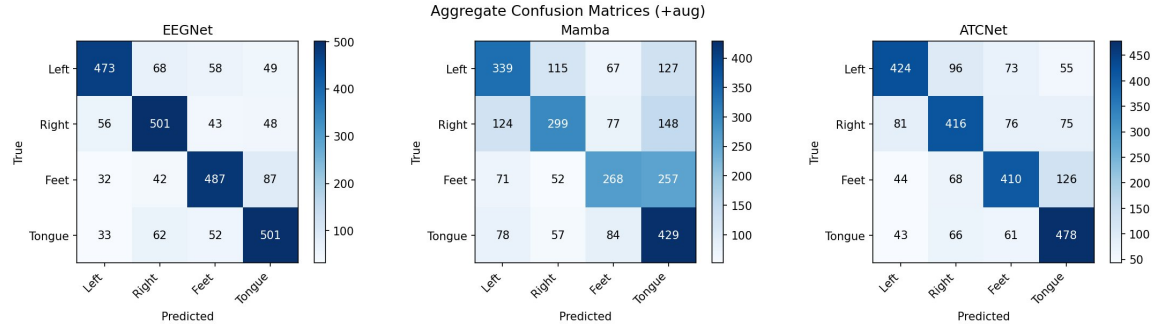


Figure 2: Aggregate confusion matrices with S&R augmentation (EEGNet, Mamba, ATCNet). FBCSP is unchanged.

Several patterns are notable. FBCSP struggles most with tongue and feet, with right hand being the best-classified class (473 correct vs 374 for tongue). This is expected: FBCSP’s spatial filtering is optimized for the

lateral ERD asymmetry of left/right hand imagery, while feet and tongue have different spatial distributions. EEGNet, ATCNet, and Mamba all classify tongue best (490, 493, and 369 correct respectively), suggesting that tongue imagery has a more distinctive neural signature that CNNs learn effectively. Mamba shows notably high confusion between feet and tongue (180 misclassifications), and its diagonal is substantially weaker than EEGNet and ATCNet, consistent with its poor overall accuracy. With augmentation, EEGNet and ATCNet maintain strong diagonals (501 and 478 tongue correct). Mamba continues to show high feet-tongue confusion (257 misclassifications) even with augmentation, confirming that the failure is architectural rather than a data quantity problem.

5.4. Spectral Analysis

Figure 3 shows the ERD% topographic maps comparing high and low performers across the mu and beta bands (8–30 Hz) during motor imagery. $ERD\% = (\text{power_imagery} - \text{power_rest}) / \text{power_rest} \times 100$; negative values indicate desynchronization (the expected motor imagery response).

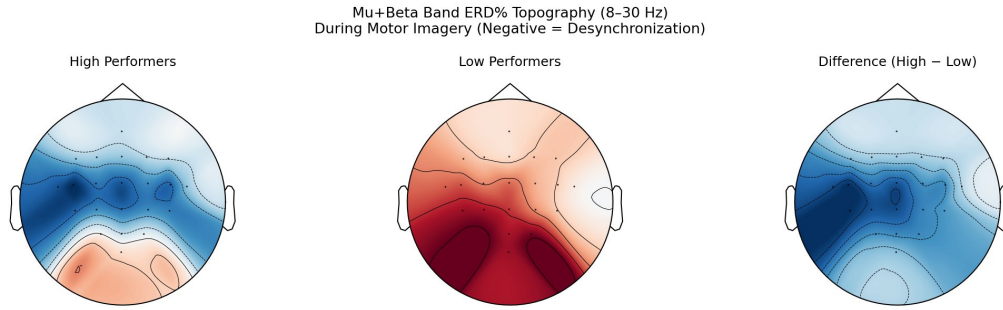


Figure 3: ERD% topographic maps (8–30 Hz) averaged across all 4 motor imagery classes and all subjects in each group. Blue = desynchronization (ERD); Red = synchronization (ERS). Left: high performers. Center: low performers. Right: difference (High – Low).

High performers (left panel) show widespread blue (desynchronization) concentrated over the central and central-posterior regions, corresponding to C3, C4, Cz, and surrounding motor cortex electrodes — the expected neurophysiological signature of motor imagery. At the channel level, high performers exhibit mu ERD% of -4.9% and beta ERD% of -6.2% at C3, indicating clear power reduction during motor imagery relative to rest.

Low performers (center panel) show the opposite: widespread red (synchronization) across most of the scalp, with mu ERD% of $+18.6\%$ and beta ERD% of -0.3% at C3. Rather than suppressing sensorimotor oscillations during motor imagery, low performers’ mu rhythms actually increase in power, while beta shows essentially no response. This means there is no ERD signal for classifiers to detect in these subjects.

The difference map (right panel) is uniformly blue, with the greatest difference over the central motor cortex region (C3/C4/Cz). This directly explains why all supervised models struggle with low performers: the feature that FBCSP, EEGNet, ATCNet, and Mamba are all trained to detect — suppression of sensorimotor rhythms — is absent in these subjects. This finding also motivates the SSL pretraining approach: if canonical ERD is not present, a model pretrained on diverse EEG from other subjects may learn to detect alternative neural patterns that supervised training on 288 trials cannot discover.

5.5. SSL-BYOL Cross-Subject Pretraining

BYOL pretraining consistently converged for all 9 subjects, with cosine similarity between online and target networks reaching approximately -0.90 by epoch 250. Table 1 shows that SSL-BYOL without augmentation achieves 68.0% mean accuracy, and SSL-BYOL + S&R achieves 73.5% — the best mean result across all conditions.

The SSL finding is most striking for subjects 05 and 06, the two subjects with most absent ERD in the topographic map. Subject 05 gains $+0.177$ over FBCSP with SSL+S&R, and subject 06 gains $+0.152$ — the two largest individual gains in the entire experiment. By contrast, subject 02, who also has absent ERD, gains only $+0.003$. This heterogeneity prompted the ablation study described below.

5.6. BYOL Pretraining Ablation

To understand which subjects’ data drives the SSL benefit for each low performer, we conducted a leave-one-out ablation: for each low-performing target subject, we re-ran pretraining 8 times, each time excluding one subject from the pretraining set (in addition to the target subject). A large accuracy drop when subject X is excluded indicates subject X ’s data was important for learning useful representations.

Figure 4 shows the accuracy drop for each excluded subject across the four low-performing targets. Two distinct patterns emerge.

Subjects 05 and 06 (most absent ERD): Subject 04 (another low performer) causes the largest drops when excluded — -0.094 for target 05 and -0.073 for target 06. Meanwhile, removing high performers 07 and 09 from target 05’s pretraining actually slightly *improves* accuracy ($+0.063$ and $+0.042$). This pattern indicates that subjects 05 and 06 benefit primarily from representations learned from other low-performing subjects’ EEG — consistent with the hypothesis that these subjects share a common alternative neural signature that BYOL captures and transfers within the low-performer group.

Subject 04: Removing high performers 07, 08, and 09 each causes a drop of -0.066 , while removing low performers causes smaller or negligible changes. This indicates subject 04 benefits primarily from canonical ERD patterns represented in high-performing subjects — a fundamentally different transfer dynamic from subjects 05 and 06.

Subject 02: Shows a mixed pattern: subject 01 (high performer) causes the largest drop (-0.066), but subjects 05 and 06 (low performers) also cause moderate drops (-0.042 and -0.052). This reflects subject 02’s consistently resistant nature across all models — it appears to benefit from both types of pretraining signal but not strongly from either.

This heterogeneity is itself a novel finding: BCI inefficiency is not a single uniform phenomenon but encompasses at least two subtypes. Subjects with completely absent ERD (05, 06) share alternative neural patterns that transfer among themselves. Subjects with weak but present ERD (04) benefit primarily from canonical high-performer representations. Subject-level ablation studies may therefore be a useful diagnostic tool for understanding which pretraining data is most informative for a given BCI-inefficient user.

[Figure: BYOL ablation bar chart — accuracy drop per excluded subject, per low-performing target. Blue bars = high performers; red bars = low performers. To be inserted from byol_ablation.png.]

Figure 4: BYOL pretraining ablation: accuracy drop when each subject is excluded from pretraining, for each low-performing target subject. Blue bars = high-performing subjects; red bars = low-performing subjects. Positive values indicate the excluded subject was a useful contributor; negative values indicate removal improved performance (interference).

6. Limitations

The primary limitation is sample size. BCI IV-2a contains only 9 subjects, yielding groups of 5 high and 4 low performers after the median split. The minimum achievable p-value with Mann-Whitney U at these group sizes is 0.095, making it mathematically impossible to reach conventional significance thresholds. All group-level findings should be interpreted as exploratory effect directions rather than confirmatory statistics.

The group split relies on FBCSP kappa as a proxy for BCI literacy, since BCI IV-2a was collected offline without real-time feedback loops. True BCI literacy is typically measured by online BCI accuracy, which captures factors including feedback adaptation and sustained attention that offline classification does not [15]. Our proxy is validated by its consistency with published per-subject rankings in the literature [9], but this limitation should be acknowledged when interpreting findings.

The dataset was collected from 9 healthy right-handed volunteers under controlled laboratory conditions. EEG characteristics of patients with motor impairments, stroke, or ALS differ substantially from healthy subjects, and direct generalization to clinical populations is not warranted without validation on appropriate datasets. No artifact rejection was applied, meaning some performance differences may reflect differences in muscular

contamination or movement artifacts rather than purely neural signal quality differences.

Finally, Mamba’s consistently poor performance may partially reflect implementation choices. The MI-Mamba architecture we implemented is our own replication from the paper description, since the original code is not publicly available [8]. Alternative implementations or hyperparameter configurations might yield different results.

7. Ethical Components

The central ethical motivation of this project is equity in assistive technology. BCI inefficiency disproportionately affects the patients who most urgently need BCIs, yet nearly all published benchmarking work reports only mean accuracy, which can mask systematic failure for a substantial minority of users. Blankertz et al. [4] reframed BCI illiteracy as a failure of the system, not the user, and our work operationalizes this framing empirically: by analyzing whether performance gains are distributed equitably across the performance spectrum, we directly ask whether the deployment of newer models would reduce or exacerbate the gap in assistive technology access. The finding that ATCNet+S&R shows the strongest differential trend toward helping low performers (+0.067 gap), while Mamba actively widens the gap (−0.081), has concrete implications for model selection in clinical contexts — deploying Mamba over FBCSP would disproportionately harm the users who already struggle most.

A meaningful limitation in this regard is that BCI IV-2a was collected from healthy volunteers. Any clinical deployment based on this research would require validation on patient populations, whose EEG characteristics may differ substantially. We emphasize that our findings should be treated as preliminary evidence motivating future validation studies, not as sufficient basis for clinical decisions. It is also worth noting that algorithmic improvements alone are insufficient for equitable BCI access — hardware cost, electrode setup time, and software usability barriers remain significant and are not addressed by model selection. Our analysis identifies which model best serves hard subjects given equivalent resources, but does not address whether those resources are equitably distributed in the first place.

8. Follow-Up Work

Short-term: The most immediate extension is adding confusion matrices stratified by high and low performer groups separately, rather than the aggregate matrices presented here. Per-group confusion patterns would reveal whether low performers show qualitatively different error distributions (e.g., more uniform confusion suggesting near-random classification) compared to high performers. Computing Cohen’s kappa alongside accuracy would also enable direct comparison to the published literature, which primarily reports kappa.

Medium-term: The ablation study revealed that subject 04 functions as an unusually informative pretraining source for other low-performing subjects. A natural follow-up is to investigate whether this reflects shared neurophysiological characteristics — for example, whether subjects 04, 05, and 06 share similar resting-state power spectra, ERD profiles, or artifact patterns that cause their EEG representations to cluster in the BYOL embedding space. This could be tested by computing pairwise cosine similarity between subjects’ representations in the pretrained encoder and correlating these similarities with pretraining contribution scores from the ablation.

Long-term: The most significant open question is whether the within-subject BCI illiteracy findings extend to cross-subject settings. A rigorous leave-one-subject-out (LOSO) evaluation would test whether models trained on other subjects can exceed FBCSP accuracy for low performers without any subject-specific calibration data — a practically critical question for real-world BCI deployment where calibration sessions are costly. Additionally, the warm-up pretraining paradigm explored here using BYOL could be extended to GAN-based generative augmentation: training a subject-conditioned generator on high-performer data and using it to synthesize plausible EEG trials for low-performer training would more directly address the absent ERD problem than S&R. Finally, validation on clinical populations (stroke, ALS, spinal cord injury) is the necessary step before any of these findings can inform clinical deployment decisions.

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